Library Management System (PS No. 99)

Test case screenshots: input and output, in order

Test Case 1: Normal Input

Input: Menu option 1 (Add book), Title: Python Basics, Author: Mark Lutz

Expected: Book added. ID: B101 | Python Basics | Mark Lutz | Available

```
===== Library Management System =====
1. Add book
2. Issue book
3. Return book
4. Search book
5. Display all books
6. Exit
Enter choice: 1
Title: python basics
Author: Mark Lutz
Book added. ID: B101 | python basics | Mark Lutz | Available
```

Test Case 2: Boundary / Minimum Input

Input: Library has only book B101; Menu option 2 (Issue): ID B101, Member: Riya

Expected: Book B101 issued to Riya. Status: Issued | Available books: 0

```
===== Library Management System =====
1. Add book
2. Issue book
3. Return book
4. Search book
5. Display all books
6. Exit
Enter choice: 2
Book ID: B101
Member: 1
Book B101 issued to 1. Status: Issued
Available books: 0
```

Test Case 3: Invalid Input

Input: Menu option 2 (Issue), Book ID: B999

Expected: Book ID B999 not found.

```
===== Library Management System =====
1. Add book
2. Issue book
3. Return book
4. Search book
5. Display all books
6. Exit
Enter choice: 2
Book ID: B999
Book ID B999 not found.
```

Test Case 4: Different Valid Scenario

Input: (a) Option 1: Data Science, Wes McKinney; (b) Option 4 (Search by author): McKinney; (c) Option 3 (Return): ID B101

Expected: (a) Book added. ID: B102 | Data Science | Wes McKinney | Available; (b) B102 | Data Science | Wes McKinney | Available; (c) Book B101 returned. Status: Available

(a) Add book

```
===== Library Management System =====
1. Add book
2. Issue book
3. Return book
4. Search book
5. Display all books
6. Exit
Enter choice: 1
Title: Data Science
Author: Wes Mckinney
Book added. ID: B102 | Data Science | Wes Mckinney | Available
```

(b) Search by author

```
===== Library Management System =====
1. Add book
2. Issue book
3. Return book
4. Search book
5. Display all books
6. Exit
Enter choice: 4
Search by (1) Title or (2) Author: 2
Enter author: Mckinney
B102 | Data Science | Wes Mckinney | Available
```

(c) Return book

```
===== Library Management System =====
1. Add book
2. Issue book
3. Return book
4. Search book
5. Display all books
6. Exit
Enter choice: 3
Book ID: B101
Book B101 returned. Status: Available
```